

## Chapter 8

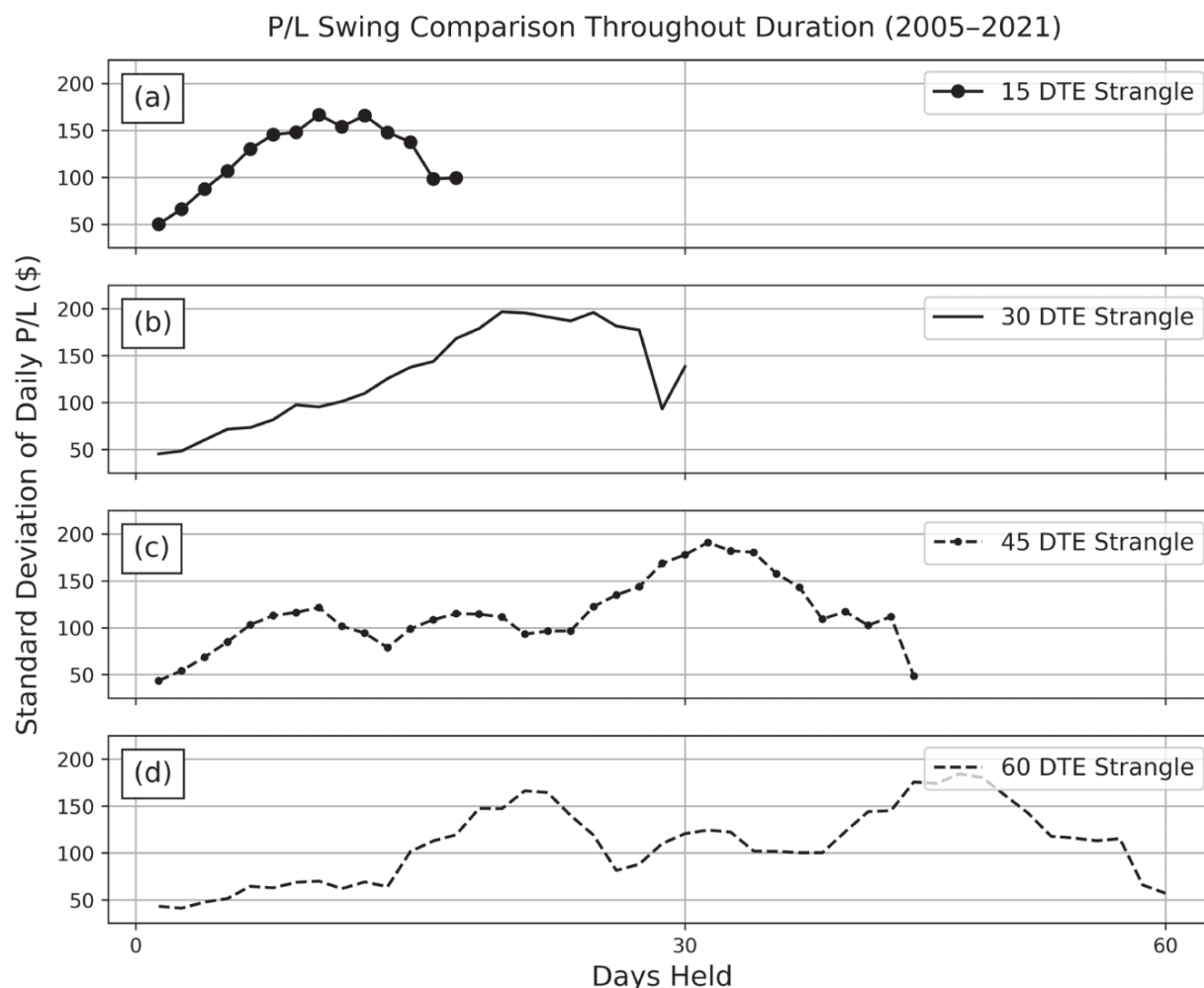
# Advanced Portfolio Management

Having covered the necessary basics of portfolio management, this chapter discusses supplemental optimization techniques for traders who can accommodate more active trading. The capital allocation guidelines, underlying diversification, and Greeks of a portfolio are essential to maintain and are relatively straightforward to employ. This chapter will introduce some less essential strategies:

- Additional option diversification techniques.
- Weighting assets according to probability of profit (POP).

## Advanced Diversification

As stated in the previous chapter, one of the biggest strategic differences between equity portfolios and options portfolios is the ability to diversify risk with respect to factors other than price. Diversifying with respect to the underlying is the most effective way to reduce the effect of outlier events on a portfolio. Diversifying with respect to other variables, such as time and strategy, requires more active management but tends to reduce the profit and loss (P/L) correlation between positions. For example, consider the per-day standard deviation of P/L for SPY strangles with different durations as shown in [Figure 8.1](#).



**Figure 8.1** Standard deviation of daily P/Ls (in dollars) for 16  $\Delta$  SPY strangles with various durations from 2005–2021. Included are durations of (a) 15 days to expiration (DTE), (b) 30 DTE, (c) 45 DTE, and (d) 60 DTE.

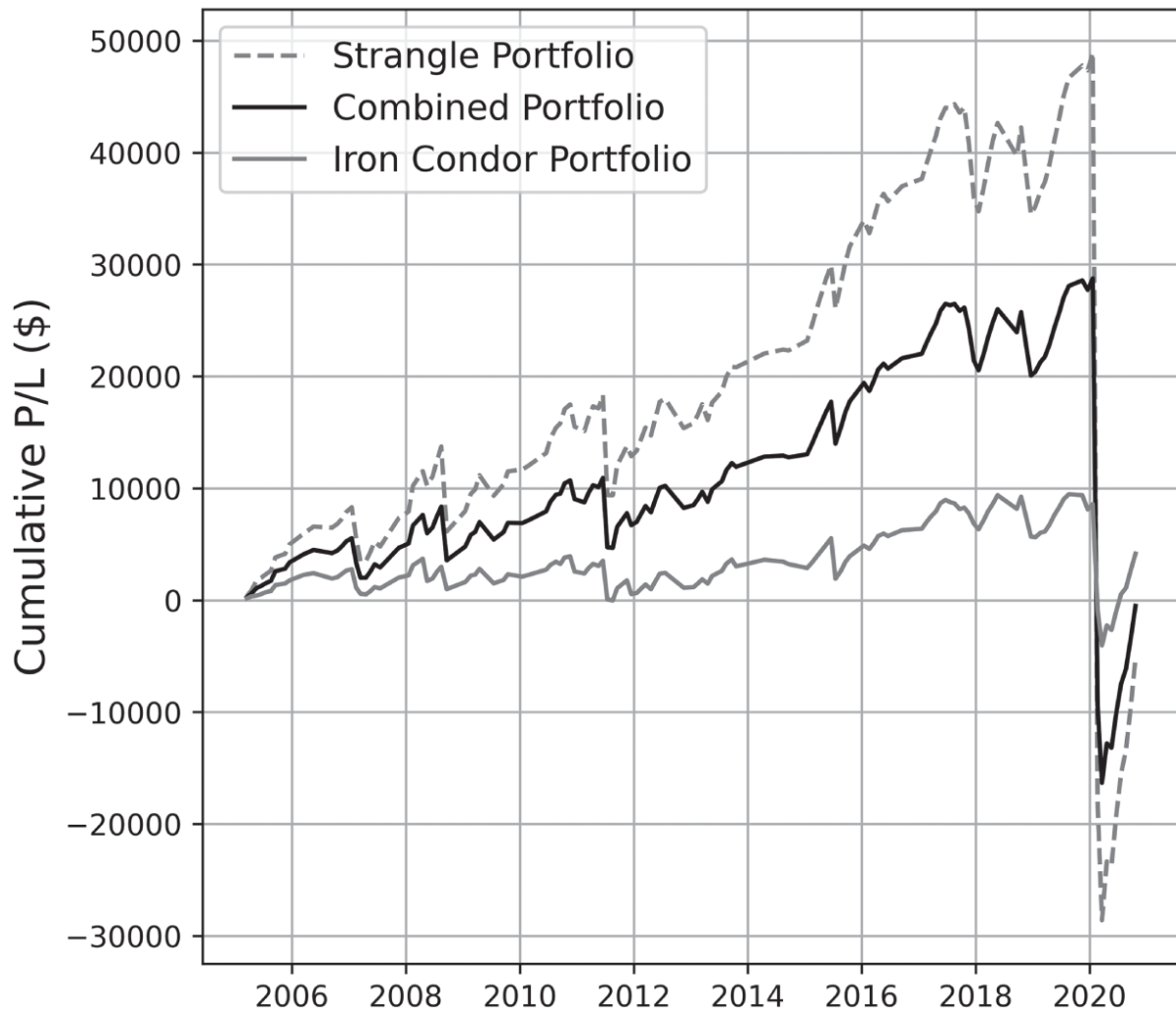
Short premium trades tend to have more volatile P/L swings as they approach expiration, a result of the position becoming more sensitive to changes in time and underlying price (larger gamma and theta). Because contracts with different durations have varying sensitivities to these factors at a given time, diversifying the timescales of portfolio positions reduces the correlations among their P/L dynamics. Because trading consistent contract durations is important for reaching many occurrences, the most effective way to diversify with respect to time is by trading contracts with consistent durations but a variety of expiration dates. This strategy achieves an assortment of contract durations in a portfolio at a given time without compromising the number of occurrences. Despite its efficacy,

diversification with respect to time will not be thoroughly covered in this chapter because it is difficult to maintain conveniently and consistently.

Strategy diversification, while not as essential as underlying diversification, is another risk management technique that is more straightforward than time diversification. This method effectively spreads portfolio capital across different risk profiles while maintaining the same directional assumption for a given underlying (or a highly correlated underlying). This lets traders capitalize on the directional dynamics of an asset while protecting a proportion of portfolio capital from outlier losses. To see an example of the diversification potential for this method, consider a backtest of three different portfolios. Each portfolio contains some combination of two directionally neutral SPY strategies: strangles and iron condors. The performance of these portfolios in this long-term backtest is shown in [Figure 8.2](#) and analyzed in [Table 8.1](#). The purpose of this backtest is not to demonstrate the profit or loss potential associated with combining SPY strangles and iron condors but rather to illustrate the possible effects of strategy diversification on portfolio risk according to one sample of outcomes.

The impact of diversification is immediately clear, particularly when emphasizing the drawdowns of the 2020 sell-off. Strangles and iron condors experienced massive drawdowns in early 2020 even though defined risk trades are lower-risk, lower-reward trades. The cumulative drawdowns as a percentage of portfolio capital are approximately the same across all three portfolios (roughly 150%). However, the drawdowns as a raw dollar amount were significantly larger for the strangle portfolio compared to the combined portfolio. During more regular market conditions, the combined portfolio also had a much larger POP and profit potential than the iron condor portfolio and less P/L variability and outlier risk than the strangle portfolio.

## Strategy Cumulative P/L Comparison (2005–2021)



**Figure 8.2** Cumulative P/L of three different portfolios containing some combination of SPY strangles and SPY iron condors, held to expiration from 2005–2021. The strangle portfolio contains 10 strangles, the combined portfolio contains five strangles and five iron condors, and the iron condor portfolio contains 10 iron condors. All contracts are traded once per expiration cycle, opened at the beginning of the expiration cycle and closed at expiration. These positions have the same short delta ( $16\Delta$ ), approximately the same duration (45 DTE), and the same open and close dates. The long strikes of the iron condors are roughly  $10\Delta$ .

This example demonstrates how diversifying portfolio capital across defined and undefined risk strategies lets a trader capitalize on the directional tendencies of an underlying asset (or several highly correlated

underlyings) while protecting a fraction of capital from unlikely outlier events. However, this example combines strategies in a highly simplified way as market implied volatility (IV), capital allocation guidelines, alternative management techniques, and strategy-specific factors are not considered. In practice, defined and undefined risk strategies reach P/L targets at different rates and often require different management strategies. The percentage of capital allocated to a single position also depends on a number of factors, including the buying power reduction (BPR) of the trade (maximum of 5% for defined risk trades and 7% for undefined risk) and the correlation with the existing positions in a portfolio. For traders interested in a more quantitative approach to positional capital allocation, allocation weights can be estimated from the probability of profit of the strategy.

**Table 8.1** Statistical analysis of the three portfolios illustrated in [Figure 8.2](#). The first four statistics (POP, average P/L, standard deviation of P/L, and conditional value at risk (CVaR)) gauge portfolio performance during more regular market conditions (2005–2020). The final column gives the worst-case drawdown from the 2020 sell-off (the cumulative losses from February to March 2020).

|                    | 2005–2020 |             |                           |           | 2020 Sell-Off        |
|--------------------|-----------|-------------|---------------------------|-----------|----------------------|
| Portfolio Type     | POP       | Average P/L | Standard Deviation of P/L | CVaR (5%) | Worst-Case Drawdowns |
| <b>Strangle</b>    | 76%       | \$379       | \$1,803                   | –\$5,174  | –\$77,520            |
| <b>Combined</b>    | 75%       | \$221       | \$1,275                   | –\$3,648  | –\$45,080            |
| <b>Iron Condor</b> | 67%       | \$64        | \$799                     | –\$2,324  | –\$12,640            |

## Balancing Capital According to POP

The proportion of capital to allocate to a position can be estimated from the POP of the strategy. An appropriate percentage of buying power can be estimated using the following formula, derived from the Kelly Criterion:<sup>1</sup>

$$f = r \cdot \frac{\text{DTE}}{365} \cdot \frac{\text{POP}}{1 - \text{POP}} \quad (8.1)$$

where  $r$  is the annualized risk-free rate of return, DTE is days to expiration or the contract duration (in calendar days), and POP is the probability of profit of the strategy.<sup>2</sup> Approximating the risk-free rate is not straightforward because it is an unobservable market-wide constant, but the long-term bond rate is commonly used as a conservative estimate. For the remainder of this chapter, the risk-free rate will be estimated at roughly 3% for the sake of simplicity. To see some examples of portfolio allocation percentages calculated using this equation, see [Table 8.2](#).

**Table 8.2** POPs and allocation percentages of buying power for 45 DTE 16  $\Delta$  SPY, QQQ, and GLD strangles from 2011–2018.

|                     | <b>Strangle Statistics (2011–2018)</b> |                               |
|---------------------|----------------------------------------|-------------------------------|
|                     | <b>POP</b>                             | <b>Allocation Percentages</b> |
| <b>SPY Strangle</b> | 79%                                    | 1.4%                          |
| <b>QQQ Strangle</b> | 73%                                    | 1.0%                          |
| <b>GLD Strangle</b> | 84%                                    | 1.9%                          |

The equation above suggests that the amount of portfolio buying power allocated to these positions should range from 1.0% to 1.9%, but those calculations don't take correlations between positions into account. Strategies with perfectly correlated underlyings should be counted against the same percentage of portfolio capital because [Equation \(8.1\)](#) requires that trades be independent of one another. In this example, because SPY and QQQ are highly correlated to each other but mutually uncorrelated with GLD, GLD strangles can occupy an entire 1.9% of portfolio buying power, and SPY strangles and QQQ strangles *combined* should occupy around 1.4% (the larger of the two allocation percentages). Because SPY and QQQ are not perfectly correlated, this is a conservative lower bound.

Overall, these allocation percentages are fairly low because the Kelly Criterion advocates for placing many small, uncorrelated bets. When aiming to allocate between 25% and 50% of portfolio buying power, strictly abiding by these bet sizes is somewhat impractical; there just aren't enough

uncorrelated underlyings. The value of the risk-free rate provides a *conservative* estimate for the ideal capital allocation, so scaling these percentages up and adopting a more aggressive approach is justified. To scale up these percentages without violating the capital allocation guidelines, these bet sizes can be used as a heuristic to estimate *proportions* of capital allocation rather than the explicit percentages. For example, rather than allocating according to POP weights, a more heuristic approach would be as follows:

- According to initial estimates, 1.4% of portfolio buying power should be allocated to SPY strangles and 1.9% to GLD strangles.
- Dividing by 1.9, these weights correspond to a ratio of approximately 0.74:1.0.
- This means that SPY strangles should occupy roughly 0.74 times the portfolio buying power of GLD strangles.
- If the maximum per-trade allocation of 7% goes toward GLD strangles, then approximately 5.2% (derived from  $0.74 \times 7\% = 5.2\%$ ) should be allocated to SPY strangles.

To continue this example, suppose that the capital allocated to SPY strangles is further split between SPY strangles and QQQ strangles. Although these underlyings are correlated, splitting capital between these positions achieves more diversification than allocating the entire 5.2% to one underlying. This process can also be estimated using the POP weights:

- According to initial estimates, 1.4% of portfolio buying power should be allocated to SPY strangles and 1.0% to QQQ strangles.
- Dividing by 2.4% (from  $1.4\% + 1.0\%$ ), these weights correspond to a ratio of approximately 0.58:0.42.
- This means that SPY strangles should occupy 58% of the capital allocation and QQQ strangles should occupy 42%.
- If a maximum of 5.2% can be allocated toward these positions, then 3.0% of portfolio capital should go toward SPY strangles and 2.2% to QQQ strangles.

This scaling formula, when combined with position sizing caps of the capital allocation guidelines, allows traders to construct portfolio weights that scale with the POP of a strategy without overexposing capital to outlier risk. These two concepts form a simple but effective basis for options portfolio construction.

## Constructing a Sample Portfolio

Throughout this section, simplified capital allocation guidelines, option diversification, and POP-weighting are combined to create a sample portfolio. The sample portfolio shown here will be constructed using data from January 2011 to January 01, 2018 and backtested with data from January 02, 2018 to September 2019. This backtest will focus on implementing some of the portfolio construction techniques outlined in [Chapters 7](#) and 8. This sample portfolio has six different core positions (all strangles), each occupying a constant amount of portfolio capital determined by the POP-weight scaling method described in the previous section. The following three simplifications are made for ease of analysis and understanding:

1. Neither market IV nor underlying IV will be considered. Scaling portfolio allocation up when market IV increases is an effective way to capitalize on higher premium prices, as is focusing on underlyings with inflated implied volatilities. Because a constant 30% of portfolio capital will be allotted to the same short premium positions throughout this backtest, profit potential will be significantly limited. Therefore, the focus of this analysis is risk management.
2. This study only uses strangles with exchange-traded fund (ETF) underlyings instead of a combination of strategies. This makes the portfolio approximately delta neutral and eliminates the need to justify specific directional assumptions or risk profiles for individual assets. By disregarding stock underlyings, stock-specific binary events, such as earnings and dividends do not apply. This also means that the added profit potential from supplemental positions (which tend to be higher risk and include stock underlyings) will not be accounted for in this backtest.



3. Rather than managing trades at fixed profit targets, all the trades shown in this backtest will be approximately opened on the first of the month and closed at the end of the month.

**Step 1:** Identify suitable underlyings using past data. Core positions should have moderate P/L standard deviations and well-diversified underlying assets. ETFs, such as the ones in [Table 8.3](#), are viable candidates for core position underlyings. Though the market ETFs are highly correlated, a sufficient number of uncorrelated and inversely correlated assets can achieve a reasonable reduction in idiosyncratic risk.

**Table 8.3** Correlations between different ETFs from 2011–2018. Included are two market ETFs (SPY, QQQ), a gold ETF (GLD), a bond ETF (TLT), a currency ETF (FXE - Euro), and a utilities ETF (XLU).

|                   |     | Correlation (2011–2018) |       |       |       |       |       |
|-------------------|-----|-------------------------|-------|-------|-------|-------|-------|
|                   |     | SPY                     | QQQ   | GLD   | TLT   | FXE   | XLU   |
| Market ETFs       | SPY | 1.0                     | 0.88  | −0.02 | −0.44 | 0.16  | 0.49  |
|                   | QQQ | 0.88                    | 1.0   | −0.03 | −0.36 | 0.12  | 0.35  |
| Diversifying ETFs | GLD | −0.02                   | −0.03 | 1.0   | 0.19  | 0.34  | 0.08  |
|                   | TLT | −0.44                   | −0.36 | 0.19  | 1.0   | −0.03 | −0.04 |
|                   | FXE | 0.16                    | 0.12  | 0.34  | −0.03 | 1.0   | 0.18  |
|                   | XLU | 0.49                    | 0.35  | 0.08  | −0.04 | 0.18  | 1.0   |

**Step 2:** Calculate the percentage of portfolio capital that should be allocated to each position. These percentages can be estimated with [Equation \(8.1\)](#) and scaled according to the methodology described in the previous section, as shown in [Table 8.4](#).

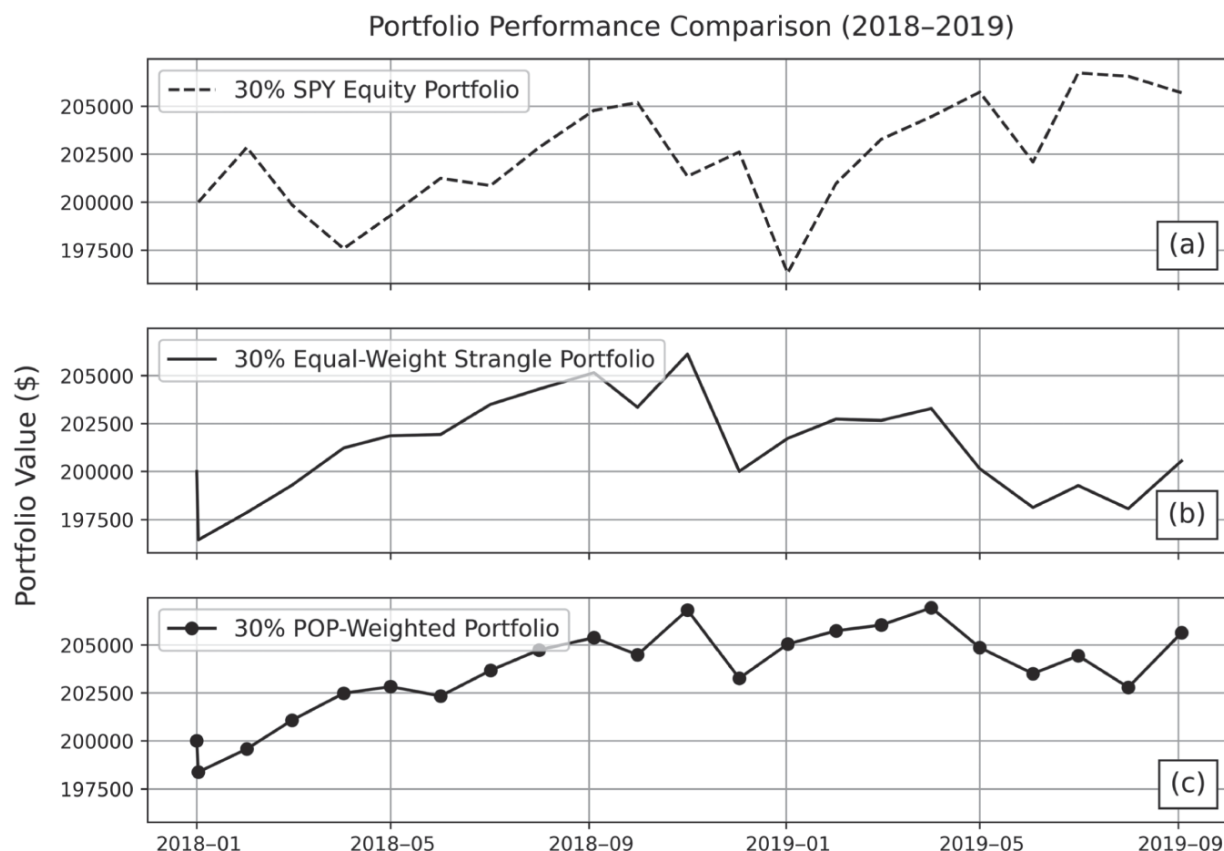
The core positions shown in [Table 8.4](#) are high-POP, have moderate P/L standard deviation, and have well-diversified underlyings, and the allocation amounts are below the 7% per-trade buying power maximum. The total portfolio buying power allocated to short premium amounts to 30%, which is close enough to the minimum 25% to suffice for this backtest. With the portfolio initialized using data from 2011 to early 2018, it can now be backtested on new data from early 2018 to late 2019, bearing in mind that this test does not take dynamic management or implied volatility

into account. The results of backtesting this sample portfolio are shown in [Figure 8.3](#) and [Table 8.5](#).<sup>3</sup>

Interestingly, [Table 8.5](#) shows that the equity portfolio was the most volatile of the three and experienced the largest worst-case drawdown despite having less tail exposure than the options portfolios. The POP-weighted portfolio performed more consistently and had significantly less per-trade standard deviation than either of the other two, with per-trade POP matching the equal-weight portfolio and average P/L comparable to the equity portfolio. Despite consisting of undefined risk strategies, the POP-weighted portfolio had nearly half the P/L variability and worst-case loss as the equity portfolio throughout the backtest period. The equal-weight strangle portfolio also underperformed compared to the POP-weighted portfolio although not experiencing any more P/L variance or severe drawdowns compared to a comparable portfolio of equities. To reiterate, the performance of both strangle portfolios can be further optimized by increasing the allocation percentage according to market volatility (which can be done with the addition of uncorrelated short premium positions) or by incorporating more complex management strategies. Still, this simplified backtest illustrates the impact of incorporating the risk management techniques of capital allocation, diversification, and POP-weighted allocation.

**Table 8.4** Core position statistics for 45 DTE 16  $\Delta$  strangles from 2011–2018. The allocation ratio is the allocation percentages normalized such that the largest bet size is set to 1.0. The portfolio weights are determined by multiplying the allocation ratio by 7% (the maximum per-trade allocation percentage). The adjusted portfolio weights show how portfolio capital is split across assets that are highly correlated.

|                                   | <b>Core Position Statistics (2011–2018)</b> |                               |
|-----------------------------------|---------------------------------------------|-------------------------------|
|                                   | <b>POP</b>                                  | <b>Allocation Percentages</b> |
| <b>SPY Strangle</b>               | 79%                                         | 1.4%                          |
| <b>QQQ Strangle</b>               | 73%                                         | 1.0%                          |
| <b>GLD Strangle</b>               | 84%                                         | 1.9%                          |
| <b>TLT Strangle</b>               | 78%                                         | 1.3%                          |
| <b>FXE Strangle</b>               | 83%                                         | 1.8%                          |
| <b>XLU Strangle</b>               | 81%                                         | 1.6%                          |
|                                   |                                             |                               |
| <b>Allocation Ratio</b>           | <b>SPY/QQQ:GLD:TLT:FXE:XLU</b>              |                               |
|                                   | <b>0.74:1.0:0.68:0.95:0.84</b>              |                               |
| <b>Portfolio Weights</b>          | <b>SPY/QQQ:GLD:TLT:FXE:XLU</b>              |                               |
|                                   | <b>5.2%:7.0%:4.8%:6.7%:5.9%</b>             |                               |
| <b>Adjusted Portfolio Weights</b> | <b>SPY:QQQ:GLD:TLT:FXE:XLU</b>              |                               |
|                                   | <b>3.0%:2.2%:7.0%:4.8%:6.7%:5.9%</b>        |                               |



**Figure 8.3** Portfolio performance of three different portfolios from early 2018 until September of 2019. Each portfolio has \$200,000 in initial capital with 30% of the portfolio capital allocated. This initial amount of \$200,000 allows at least one trade for each type of position, as \$100,000 in initial capital does not. The 30% SPY equity portfolio (a) has 30% allocated to shares of SPY. The 30% equally-weighted strangle portfolio (b) has 5% allocated to each of the six types of strangles, and the 30% POP-weighted portfolio (c) has the 30% weighted according to the percentages in [Table 8.4](#). All contracts have the same delta (16  $\Delta$ ), identical durations (roughly 45 DTE), and the same open and close dates. For the sake of comparison, the trades in the equity portfolio are opened on the first of each month and closed at the end of each month.

**Table 8.5** Portfolio backtest performance statistics for the three portfolios described in [Figure 8.3](#) from 2018–2019.

|                       | <b>Portfolio Performance Comparison (2018–2019)</b> |                    |                                  |                   |
|-----------------------|-----------------------------------------------------|--------------------|----------------------------------|-------------------|
| <b>Portfolio Type</b> | <b>POP</b>                                          | <b>Average P/L</b> | <b>Standard Deviation of P/L</b> | <b>Worst Loss</b> |
| <b>SPY Equity</b>     | 60%                                                 | \$285              | \$2,879                          | –\$6,319          |
| <b>Equal-Weight</b>   | 67%                                                 | \$26               | \$2,440                          | –\$6,117          |
| <b>POP-Weighted</b>   | 67%                                                 | \$268              | \$1,610                          | –\$3,561          |

The  $\frac{\text{POP}}{1-\text{POP}}$  heuristic derived from the Kelly Criterion provides a good guide for how much capital should be allocated to a trade when initializing a portfolio, indicating that more capital should be allocated to higher POP trades and less capital should be allocated to less reliable trades. However, this method does not provide a thorough structure for dynamic portfolio management. At different points in time, trades often reach profit or loss targets, require strike re-centering, or present new opportunities. Traders can simplify the complex management process by, for example, choosing the same contract duration or management strategy for all trades in a portfolio. However, a framework for navigating these dynamic circumstances is still necessary, and this is where the portfolio Greeks and the re-balancing protocol outlined in [Chapter 7](#) are particularly useful.

## Takeaways

1. Options can be diversified with respect to a number of variables, but diversifying the equity underlyings of an options portfolio remains the most essential for portfolio risk management. Traders who can accommodate more involvement and are interested in further diversification can also diversify positions with respect to time and strategy.
2. Diversification with respect to time tends to reduce the correlations between portfolio positions because contracts respond differently to changes in time, volatility, and underlying price depending on their

duration. The most effective way to diversify with respect to time without compromising occurrences is by trading contracts with consistent durations but a variety of expiration dates. This strategy is difficult to maintain consistently, however, particularly when multiple management strategies are used.

3. Diversifying portfolio capital across defined and undefined risk strategies allows traders to capitalize on the directional tendencies of an underlying asset while protecting a fraction of capital from unlikely outlier events. If implementing this diversification technique, note that defined and undefined risk strategies typically reach P/L targets at different rates and often require different management strategies.
4. The percentage of capital allocated to a single position can be calculated from the POP of the strategy and the correlation between existing portfolio positions. The percentage of portfolio capital allocated to a single position can be estimated using [Equation \(8.1\)](#); however, this percentage can also be scaled up because the risk-free rate yields a very conservative estimate.

## Notes

- [1](#) For an introduction to the Kelly Criterion, refer to the appendix.
- [2](#) The POPs used throughout this chapter are calculated from historic options data. Options data are ideal for statistical analyses but inaccessible to most people. Trading platforms often provide the theoretical POP of a strategy, which can substitute measured POP for these calculations.
- [3](#) This backtest demonstrates one specific outcome out of many possible when trading short premium. The goal of this backtest is to demonstrate how one sample portfolio performs relative to other portfolios with similar characteristics under these specific circumstances.